

# AP CSP | Lesson 20: Searching Algorithms

## Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

### Questions 1-6 | 1 mark each

#### Q1. Answer: A

A uses AP one-based indexing. B uses a zero-based interpretation. C selects the last element. D returns the index rather than the stored value.

#### Q2. Answer: B

B guarantees that the left values are no larger than the midpoint. Positivity and odd length are not prerequisites. A target may be absent; binary search must still conclude not found.

#### Q3. Answer: C

C counts 3,8,12,19,27,34. A matches binary search for this target. B stops before examining the match. D unnecessarily examines the last element after finding it.

#### Q4. Answer: D

D keeps values strictly above the checked 19. A keeps the wrong half. B retains an already ruled-out midpoint. C assumes the answer before inspecting the next midpoint.

#### Q5. Answer: B and C

B stops after all candidates are eliminated. C separates correctness from cost. A is false because linear search can inspect unsorted data. D fails when the target is the first element of the list. Award one mark only for B and C together.

#### Q6. Answer: B

B counts the five nonempty intervals. A counts gaps between them instead of checks. C is the size after one elimination. D is the linear-search maximum.

### Written-response answers | Questions 7-12

#### Q7. 4 marks

**Answer:** 19 → 34 → 50; 3 checks.

**1 mark:** Gives first inspected value 19.

**1 mark:** Gives second inspected value 34.

**1 mark:** Gives third inspected value 50.

**1 mark:** Gives three element checks.

#### Q8. 4 marks

**Answer:** 19 → 34 → 27; false because after ruling out 27 no candidate remains.

**1 mark:** Gives first inspected value 19.

**1 mark:** Gives second inspected value 34.

**1 mark:** Gives third inspected value 27.

**1 mark:** Explains that all candidates are eliminated without a match, so the result is false.

**Q9. 4 marks**

**Answer:** The data are unsorted. Remaining values are 3 and 5. The target 9 was discarded on the left. Repair: sort first and apply binary search, or use linear search on the unsorted list.

**1 mark:** Identifies the missing sorted-order prerequisite.

**1 mark:** Gives remaining values 3 and 5.

**1 mark:** Explains that the actual 9 on the left has been discarded.

**1 mark:** Gives a valid repair: linear search directly or sort before binary search.

**Q10. 4 marks**

**Answer:** Linear: 1 check. Binary: 3 checks (19,8,3). Both return true. Linear checks fewer elements on this valid input, so the universal claim is false.

**1 mark:** Gives linear count 1.

**1 mark:** Gives binary count 3.

**1 mark:** States that both return true for membership.

**1 mark:** Uses this input as a counterexample to the claim that binary always takes fewer checks.

**Q11. 4 marks**

**Answer:** Linear returns true on a matching element; false after checking every element without a match. Binary returns false after no candidate remains without a match. Agreement means both return the same membership Boolean for that list and target.

**1 mark:** States that a matching element makes linear search succeed.

**1 mark:** States that checking the whole list without a match makes linear search fail.

**1 mark:** States that exhausting binary candidates without a match makes binary search fail.

**1 mark:** Defines agreement as the same true/false membership result for the same valid input.

**Q12. 4 marks**

**Answer:** One unsorted query: linear search is a simple valid choice because it requires no sorting. Many queries on already sorted data: binary search can reduce candidates by about half per step. Sorting has a cost if it is needed; compare preparation plus searches when choosing for unsorted data.

**1 mark:** Recommends linear search as a suitable option for the one-time unsorted query.

**1 mark:** Explains that linear search works without sorting or examines elements directly.

**1 mark:** Recommends binary search for already sorted repeated queries and relates it to interval halving.

**1 mark:** Acknowledges sorting/preparation cost when evaluating a strategy for initially unsorted data.